
A POLAR COORDINATE PRIOR FOR SELF-SUPERVISED STRONG-LENS REPRESENTATIONS

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ABSTRACT

We introduce a polar coordinate prior that improves self-supervised representations for dark matter substructure classification in simulated strong gravitational lenses. The prior combines polar resampling about the lens centre with an azimuthal residual channel, obtained by subtracting the mean intensity at each radius. This fixed input transform exposes non-axisymmetric structure while preserving the encoder backbone and training objectives. On DeepLense simulations, we compare Cartesian, polar-only and polar-plus-residual inputs under I-JEPA and masked autoencoders, holding the backbone, token count, masking and training schedule fixed. The full prior improves frozen-feature accuracy for three-class classification by 10–25 percentage points across label budgets of 1%, 10% and 100%, with gains in all 18 paired seed comparisons. The residual improves mean accuracy beyond polar resampling alone at every tested label budget under both objectives. The same prior reduces test errors by factors of 2.5–4.2 across five supervised networks. These gains require no learned preprocessing and increase the self-supervised encoder’s parameter count by only 0.8%. The results establish the polar prior as a practical way to incorporate lens geometry into both self-supervised and supervised learning.

1 INTRODUCTION

Strong gravitational lensing is one of the few ways to detect dark matter on scales smaller than galaxies (Vegetti et al., 2012; Hezaveh et al., 2016). A massive foreground galaxy bends the light of a background source into arcs, often close to a complete Einstein ring. Most of what we see is set by the smooth mass distribution of the lens. Substructure in its dark matter halo adds small perturbations on top of that: cold dark matter predicts compact subhaloes, while ultralight axion-like dark matter produces wave-like interference patterns (Hu et al., 2000). Telling the two apart from images means finding small, localised departures from a bright ring that is otherwise close to axisymmetric.

Neural networks trained on simulations can separate these substructure classes (Alexander et al., 2020a; Diaz Rivero and Dvorkin, 2020), and recent work builds the physics of lensing into the model, through physics-informed architectures (Veloso et al., 2023; Ojha et al., 2024) or lensing-aware self-supervised methods (Chang et al., 2023; Rishi et al., 2025). Self-supervised pretraining suits the problem because lens images are plentiful while trusted labels are not. Upcoming surveys will find many thousands of lenses, and none of them will arrive labelled with a dark matter model. The standard objectives, however, were developed for natural photographs. I-JEPA (Assran et al., 2023) and masked autoencoders (MAE; He et al., 2022) treat a lens image like any other grid of pixels, with no notion that it has a centre, or that rotating it about that centre changes nothing physical.

A lens has both. Its centre is a natural origin, and rotation about the centre is a natural symmetry. In polar coordinates about that point, a smooth Einstein ring becomes a horizontal band that hardly varies with angle, and a rotation of the image becomes a cyclic shift along the angular axis. Subtracting the mean intensity at each radius then removes the axisymmetric part altogether and leaves the perturbations that the substructure classes differ by. We call the two-channel input formed by the polar image and this azimuthal residual the polar prior (Figure 1).

Polar coordinates are not new to computer vision (Esteves et al., 2018), and they have recently been applied to lens images: Campbell et al. (2026) showed that polar inputs help a supervised convolutional network detect subhaloes in simulated Hubble Space Telescope observations. Equivariant networks instead build rotational symmetry into the architecture itself (Cohen and Welling, 2016; Weiler and Cesa, 2019). Our question concerns self-supervision. Does a change of coordinates, applied in front of a standard encoder, help when the representation is learned without labels, and what does the azimuthal residual add beyond the resampling?

We study this with a controlled comparison inside two self-supervised objectives in which only the input changes. The encoder, token count, masking distribution and optimisation schedule are identical across arms; only the number of input channels differs. Our contributions are:

- A drop-in input prior for self-supervised lensing. Polar resampling with an azimuthal residual, placed in front of an unchanged vision transformer, raises frozen-feature accuracy by 10 to 25 points under both I-JEPA and MAE, and wins every one of 18 paired seed comparisons.
- A three-way comparison that separates coordinates from residual. Polar-only input already improves on Cartesian input, and adding the residual as a second channel improves on polar-only at every label fraction for both objectives.
- Transfer beyond self-supervision. The same input reduces test errors by factors of 2.5 to 4.2 across five supervised networks that span two orders of magnitude in size.

1 Sample about the lens centre

2 Form two complementary channels

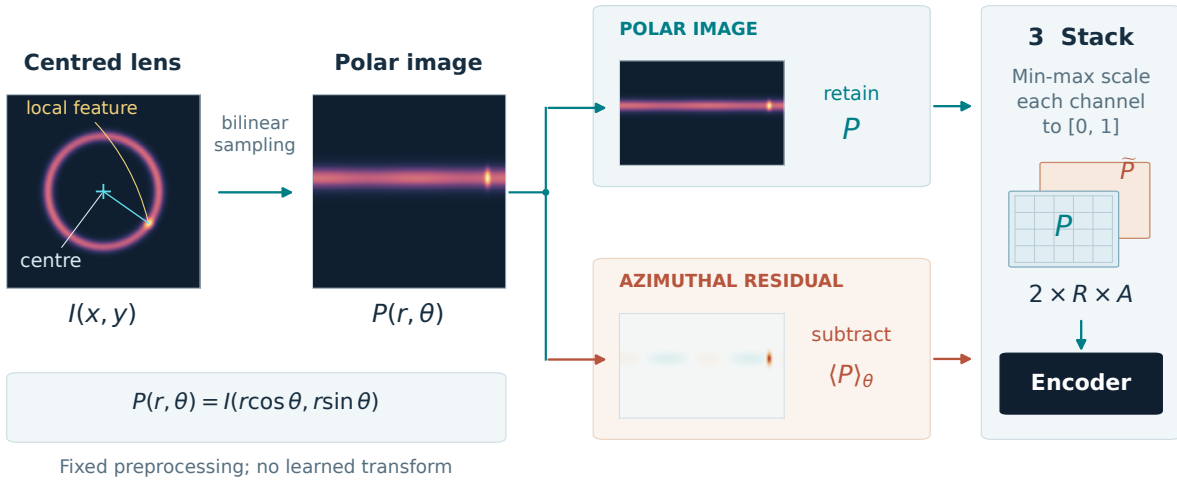


Figure 1: The polar input prior, illustrated on a synthetic lens. Bilinear resampling about the lens centre turns the ring into a band. Subtracting the angular mean at each radius removes the axisymmetric component and exposes local deviations. The polar image is retained alongside this signed azimuthal residual; each channel is independently min-max scaled to $[0, 1]$ before stacking as the encoder input. Residual colours show negative and positive deviations before scaling. No evaluation image is shown.

2 RELATED WORK

Substructure from lens images. Machine learning on this problem began with supervised classifiers trained on simulated lenses (Alexander et al., 2020a; Diaz Rivero and Dvorkin, 2020), followed by unsupervised approaches (Alexander et al., 2020b). Later models build the lens equation into the network, through a branch that approximates the lensing potential (Veloso et al., 2023) or through physics-informed tokenisation (Ojha et al., 2024). These methods change the architecture. The polar prior changes only the input, so it can be used alongside any of them.

Self-supervised learning for lenses. Self-supervised representations have been used to search survey images for new lens candidates (Stein et al., 2022). LenSiam (Chang et al., 2023) adapts contrastive pretraining with augmentations designed for lenses, and Lens-JEPA (Rishi et al., 2025) adds physics guidance to I-JEPA. Both modify the pretraining method. We keep the pretraining objectives fixed and change only what they are given to look at.

Symmetry and coordinates. Group-equivariant and steerable networks enforce rotational symmetry exactly (Cohen and Welling, 2016; Weiler and Cesa, 2019), at a compute cost that grows with the size of the group. Polar transformer networks (Esteves et al., 2018) learn an origin and resample the image in log-polar coordinates around it, which turns rotation and scaling about that origin into translations. Our origin is not learned; it is fixed by the lens. For lenses specifically, Campbell et al. (2026) apply a polar transform before a supervised convolutional network and report roughly 15% higher detection rates for subhaloes of 10^9 to $10^{9.5}$ solar masses, with the largest gains at low signal-to-noise. We share their change of coordinates, but we add the azimuthal residual and study self-supervised pretraining for classifying dark matter models rather than supervised subhalo detection.

3 STRONG LENSES AND THE SIMULATIONS

We use the public DeepLense simulations, generated with DeepLenseSim (Toomey) on top of lenstronomy (Birrer and Amara, 2018; Birrer et al., 2021). They contain three balanced classes, following Alexander et al. (2020a): lenses with cold dark matter subhaloes, lenses with axion-like vortex substructure, and lenses with no substructure. The classes differ only in what is added to the main halo, so the signal is a perturbation of the arcs rather than a change in the overall ring.

We use two simulation sets. Model I images are 150×150 pixels, with a Gaussian point spread function and Gaussian and Poisson noise at a signal-to-noise ratio of about 25. We pretrain on 9,999 of them and never use their labels. Model II images are 64×64 pixels with Euclid-like observing characteristics. All downstream evaluation uses Model II, and no Model II image is seen during pretraining, so every downstream result measures transfer to a different simulated instrument.

Two properties of these images drive our design. The lens sits at the centre of the frame, and the point spread function is isotropic. The centre is therefore the natural origin for a coordinate system, and rotating the image about it produces an equally valid observation of the same system, with the same class. We take the centre as known, as it is in the simulations; for survey images it would come from a fit to the lens galaxy.

4 THE POLAR PRIOR

Let $I(x, y)$ be an image with the lens centre at the origin. We sample it on R radii and A angles,

$$P(r_i, \theta_j) = I(r_i \cos \theta_j, r_i \sin \theta_j), \quad i = 0, \dots, R-1, \quad j = 0, \dots, A-1, \quad (1)$$

using bilinear interpolation, with radii spaced evenly from the centre to the edge of the frame and angles spaced evenly around the circle. The azimuthal residual subtracts the mean over angle at each radius,

$$\tilde{P}(r_i, \theta_j) = P(r_i, \theta_j) - \frac{1}{A} \sum_{k=0}^{A-1} P(r_i, \theta_k). \quad (2)$$

The encoder receives the two-channel tensor $[\text{mm}(P), \text{mm}(\tilde{P})]$, where $\text{mm}(\cdot)$ rescales each channel of each image to the range $[0, 1]$ using that channel’s minimum and maximum. We refer to this input as polar+residual, and to the single channel $\text{mm}(P)$ as polar-only. Cartesian inputs are likewise min-max scaled per image to $[0, 1]$.

Three properties follow from the construction; Figure 1 illustrates the construction on a synthetic lens. First, rotating the image about its centre by $2\pi m/A$ shifts both channels cyclically by m columns. The rotational symmetry of the lens becomes a translation along a single axis. This does not make a network rotation-invariant on its own, since the network must also be insensitive to the shift, but it moves the symmetry into a much simpler form. Second, any part of the image that does not vary with angle is removed exactly by Eq. (2), so to the extent that the smooth lens is axisymmetric, $\tilde{P}$ contains only the departures from it. Third, the transform is cheap: one resampling and one average per image, with nothing learned. In our transformer the only architectural change is a patch embedding that reads two channels instead of one (and, for MAE, a decoder that predicts both), which increases the encoder’s parameter count by 0.8%, from 21.52M to 21.69M.

5 EXPERIMENTAL SETUP

Self-supervised pretraining. Every arm uses the same vision transformer (ViT): three blocks of width 768, with 15×15 -pixel patches that turn a 150×150 input into 100 tokens. Model II images are resized to 150×150 before

any further processing, and the polar grid has 150 radii and 150 angles. We pretrain on the 9,999 Model I images for 300 epochs with AdamW and a batch size of 32, using the same learning-rate and weight-decay schedules for every arm (Appendix A). We use two objectives. I-JEPA predicts the representations of masked target blocks from a visible context block, with targets supplied by an exponential-moving-average copy of the encoder; our implementation adds variance and covariance penalties in the style of VICReg (Bardes et al., 2022). MAE reconstructs the pixels of randomly masked patches, here 75% of them. Within each objective the Cartesian, polar-only and polar+residual arms differ only in their input, and each arm is trained with three seeds, for 18 pretraining runs in total.

Frozen evaluation. We take 3,000 Model II images per class and fix a single 80/20 split into 7,200 training and 1,800 test images. For each pretrained encoder we concatenate the mean- and max-pooled output tokens, standardise the features, and fit a logistic regression on 1%, 10% or 100% of the training images, that is 72, 720 or 7,200 examples, with subsets stratified by class. We report three-class test accuracy as the mean and sample standard deviation over the three pretraining seeds. At full labels we also report the macro-averaged one-vs-rest area under the ROC curve (ROC AUC).

Supervised check. To see whether the prior depends on self-supervision, we also train five networks end to end on Cartesian and polar+residual input. They span two orders of magnitude in size: a small three-block convolutional network (BasicCNN), two ordinary-convolution transcriptions of a steerable-CNN topology (PlainNet-P and PlainNet-W), a vision transformer designed for small datasets (ViT-SD; Lee et al., 2021), and ResNet-18 (He et al., 2016). None is equivariant, and none uses circular padding or a circular positional encoding, so nothing in the architectures is adapted to the polar input. Each network is trained on 9,000 Model II images, with 5,400 for validation, and tested on a fixed set of 30,000. Learning rates were tuned on Cartesian input and kept unchanged for the polar runs. Checkpoints are chosen by validation macro-averaged F1 score, and every configuration is run with three seeds.

6 RESULTS

6.1 The prior improves both objectives at every label fraction

Table 1 and Figure 2 summarise the frozen evaluation. Polar+residual input improves on Cartesian input in all six combinations of objective and label fraction. For I-JEPA the gains are 21.0, 20.7 and 15.7 percentage points at 1%, 10% and 100% of the labels; for MAE they are 24.7, 15.9 and 10.4. The improvement holds run by run, not only on average: in each of the 18 comparisons between seed-matched encoders (Appendix B), the polar+residual encoder beats its Cartesian counterpart. At full labels the ROC AUC rises from 0.904 to 0.985 for I-JEPA and from 0.944 to 0.986 for MAE.

The largest gains come where labels are scarcest. With 72 labelled images, Cartesian I-JEPA features are barely above chance, at 0.365 against 0.333, whereas the same objective on polar+residual input reaches 0.576. MAE starts from stronger Cartesian features, 0.428, and climbs further, to 0.675. One of the three MAE polar+residual seeds is weaker than the other two, which gives that arm the largest spread in Table 1, but it still beats its paired Cartesian seed at every label fraction.

Table 1: Frozen three-class accuracy on Model II (mean $\pm$ sample standard deviation over three pretraining seeds). Polar-only isolates the change of coordinates; polar+residual is the full prior.

Objective	Input	1% labels	10% labels	100% labels
I-JEPA	Cartesian	0.365 $\pm$ 0.018	0.577 $\pm$ 0.043	0.771 $\pm$ 0.019
	Polar-only	0.448 $\pm$ 0.078	0.632 $\pm$ 0.083	0.801 $\pm$ 0.068
	Polar+residual	0.576 $\pm$ 0.017	0.784 $\pm$ 0.019	0.928 $\pm$ 0.006
MAE	Cartesian	0.428 $\pm$ 0.010	0.709 $\pm$ 0.023	0.832 $\pm$ 0.010
	Polar-only	0.540 $\pm$ 0.044	0.809 $\pm$ 0.010	0.878 $\pm$ 0.009
	Polar+residual	0.675 $\pm$ 0.137	0.868 $\pm$ 0.097	0.936 $\pm$ 0.061

6.2 Coordinates and residual both contribute

The polar-only arm isolates the change of coordinates, and on its own it already helps: relative to Cartesian input it adds 8.3, 5.5 and 3.0 points for I-JEPA and 11.2, 10.0 and 4.6 points for MAE. Adding the residual as a second channel contributes a further 12.8, 15.2 and 12.7 points for I-JEPA and 13.5, 6.0 and 5.8 points for MAE, so polar+residual has the higher mean at every label fraction under both objectives. For individual seeds, all nine I-JEPA comparisons

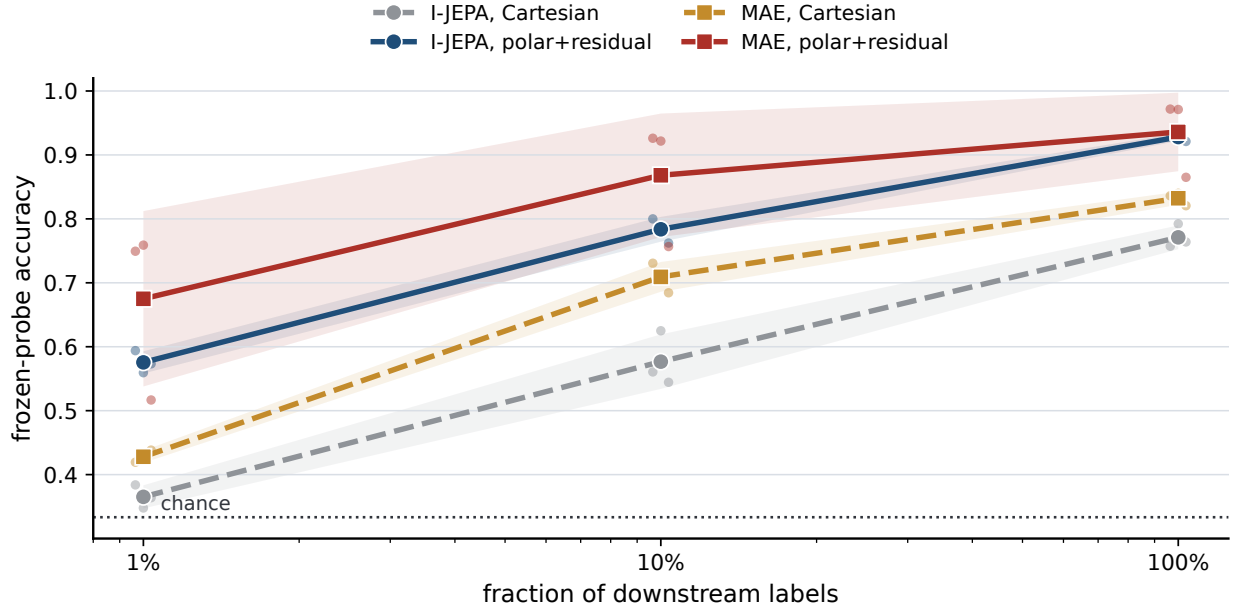


Figure 2: Frozen-feature accuracy against the fraction of downstream labels. Lines and large markers show the mean over three pretraining seeds, shaded bands one sample standard deviation, and small points the individual seeds. The dotted line marks chance. Polar+residual improves on Cartesian input for both objectives at every fraction.

favour polar+residual over polar-only, as do six of the nine MAE comparisons; the three exceptions all involve the weaker MAE seed mentioned above.

6.3 The gain carries over to supervised training

Table 2 reports the supervised check. Polar+residual input lowers the number of test errors on every network, by factors between 2.5 and 4.2. The reduction is about as large for the 0.1M-parameter BasicCNN as for the 11M-parameter ResNet-18, which argues against a regularisation effect that only helps small models. It also persists on networks that are already very accurate: PlainNet-W improves from 99.41% to 99.80%, with 2.9 times fewer errors. These gains are accompanied by modest increases in parameter count (at most 0.25% for the CNNs and 1.81% for ViT-SD), which may contribute in part to the improvements. Because the supervised runs follow a different protocol from the frozen evaluation, we do not compare the two sets of numbers directly. Their purpose is to show that the benefit is not tied to one encoder or one pretraining objective.

Table 2: Supervised accuracy on 30,000 held-out Model II images, averaged over three seeds. Error reduction is the ratio of Cartesian to polar+residual test errors.

Network	Parameters	Cartesian	Polar+residual	Error reduction
BasicCNN	0.107M	0.9060	0.9710	3.24×
PlainNet-P	0.108M	0.9918	0.9977	3.55×
PlainNet-W	1.284M	0.9941	0.9980	2.90×
ViT-SD	4.1M	0.9214	0.9691	2.54×
ResNet-18	11.2M	0.9730	0.9936	4.22×

7 DISCUSSION AND LIMITATIONS

The polar prior asks very little of a pipeline. It is a fixed function of the image, it adds no learned components beyond one extra input channel, and it leaves the encoder and the objective alone. That also means it complements the approaches in Section 2 rather than competing with them: a physics-informed architecture or a lensing-aware objective can take polar input as easily as Cartesian input.

What the prior does need is the lens centre. Both channels are defined about it, so the prior’s accuracy depends strongly on how well the centre is known. Moving the origin away from the true centre makes the radius of the ring vary with angle, which appears in the residual as a dipole pattern on the scale of the ring rather than of the substructure. Applying the prior to survey images will therefore require an accurate centre estimate, or training that exposes the network to uncertainty in the centre.

Natural next steps are lens finding in survey data, where cutouts are already centred on candidate lens galaxies, and other downstream tasks such as super-resolution.

8 CONCLUSION

We have shown that a polar coordinate prior improves both self-supervised and supervised learning for dark matter substructure classification in simulated strong lenses. Combining polar resampling with an azimuthal residual increases frozen-feature accuracy by 10–25 percentage points under I-JEPA and MAE across all three tested label budgets, with the largest gains when labels are scarce. Ablations show that the residual provides additional benefit beyond polar resampling alone. The same prior reduces test errors by factors of 2.5–4.2 across five supervised networks. These improvements require no learned preprocessing, preserve the encoder backbone and training objectives, and increase the self-supervised encoder’s parameter count by only 0.8%. Together, these results demonstrate the value of incorporating lens geometry directly into the input representation, providing a practical prior that strengthens learning across different objectives and architectures.

APPENDICES

A IMPLEMENTATION DETAILS

Polar grids. In the self-supervised experiments, images are resized to 150×150 pixels and resampled on 150 radii and 150 angles. The angular samples include both ends of $[0, 2\pi]$, so the first and last columns of the polar image coincide; every self-supervised arm uses the same grid. In the supervised experiments the grid has 64 radii and 128 angles at $\theta_j = 2\pi j/A$, sampled from the native 64×64 Model II images. In both cases the radii run from the centre to the edge of the frame, and resampling is bilinear.

Pretraining. AdamW with batch size 32 for 300 epochs. The learning rate warms up from 2×10^{-4} to 10^{-3} over 40 epochs and then decays to 10^{-6} ; weight decay increases from 0.04 to 0.4. The I-JEPA target encoder is an exponential moving average of the context encoder, with momentum increasing from 0.996 to 1, and the VICReg-style variance and covariance penalties have weights 0.25 and 0.04. MAE masks 75% of patches and, for two-channel input, reconstructs both channels. Apart from the patch embedding and the MAE reconstruction target, the arms within an objective are identical.

Frozen probe. Features are the concatenated mean- and max-pooled output tokens, standardised before fitting. The classifier is scikit-learn multinomial logistic regression with the L-BFGS solver, L2 regularisation and inverse regularisation strength $C = 1.0$, with a maximum of 5,000 iterations. These settings are fixed across input arms and label fractions. The 1% and 10% subsets are drawn separately from the training partition, stratified by class, using random seed 42.

Supervised training. Batch size 64, with no weight decay on normalisation and bias parameters. Learning rates were tuned per network on Cartesian input and fixed before any polar run. Checkpoints are selected by validation macro-averaged F1; no selection, stopping or tuning decision used the test set.

Supervised networks. BasicCNN has three convolution–ReLU–max-pool blocks followed by a flattened linear classifier. PlainNet-P and PlainNet-W reproduce the eight-layer topology of a C8-steerable CNN (Weiler and Cesa, 2019) with ordinary convolutions, batch normalisation and global average pooling; PlainNet-P matches the steerable network’s parameter count and PlainNet-W its channel widths. ViT-SD uses shifted-patch tokenisation and locality self-attention (Lee et al., 2021). ResNet-18 is the standard architecture (He et al., 2016). Parameter counts are listed in Table 2.

B SEED-LEVEL RESULTS

Table 3: Frozen accuracy for each pretraining seed, listed in the same seed order for every arm.

Objective	Input	1%	10%	100%
I-JEPA	Cartesian	.384 / .348 / .364	.561 / .625 / .544	.757 / .792 / .764
	Polar-only	.429 / .381 / .533	.598 / .571 / .727	.783 / .743 / .876
	Polar+residual	.594 / .559 / .573	.800 / .789 / .762	.932 / .931 / .921
MAE	Cartesian	.419 / .426 / .438	.731 / .713 / .684	.836 / .840 / .821
	Polar-only	.491 / .575 / .554	.802 / .819 / .804	.868 / .886 / .879
	Polar+residual	.749 / .759 / .517	.926 / .922 / .757	.972 / .971 / .865

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